

NARGES GHASEMI

University of Southern California, Los Angeles, CA

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Education

University of Southern California

Ph.D. in Computer Science; GPA: 3.96/4.0

Aug 2023 – May 2029 (Expected)

Los Angeles, CA

Shahid Beheshti University

B.Sc. in Computer Engineering; GPA: 3.96/4.0; ranked 1st of 131

Sep 2018 – Feb 2023

Tehran, Iran

Farzanegan1 High School

Diploma in Mathematics and Physics; GPA: 4.0/4.0

Sep 2014 – Jun 2018

Tehran, Iran

National Organization for the Development of Exceptional Talents (NODET)

Research Focus

Low-resource language model adaptation, multilingual NLP, language model steering and interpretability, and autoregressive modeling for computer vision.

Research and Professional Experience

NLP Researcher

USC Information Sciences Institute, CUTE LABNAME

Advisor: Prof. Jonathan May

Jul 2025 – Present

Marina del Rey, CA

- **Low-resource language model adaptation:** Adapting open-weight LLMs for low-resource languages through synthetic-data generation, continued pretraining, and translation and sentiment fine-tuning as part of DARPA-funded research.
- Built an end-to-end synthetic-data generation and validation pipeline; improved translation quality by combining complementary machine translators and related-language supervision.
- **ORBIT:** Developed training-free multi-attribute steering through a joint activation subspace and a single norm-preserving rotation; evaluated behavioral control and coherence on Llama and Qwen models.
- **Chain-of-thought faithfulness:** Developing a causal evaluation pipeline combining sentence-level counterfactuals, sparse-feature attribution, and interventions to test whether reasoning features affect final answers.
- **Bias and deception:** Investigating emotion-related activation vectors to detect and analyze deceptive and biased behavior in language models.

CV Researcher

InfoLab, University of Southern California

Aug 2023 – Jul 2025

Los Angeles, CA

- **GeoToken (ICDM 2025):** Co-developed image geolocalization inspired by **autoregressive language modeling**, representing locations as hierarchical token sequences and predicting them token by token conditioned on an image.
- **OSMGen (NeurIPS UrbanAI Workshop 2025):** Co-developed controllable satellite image synthesis conditioned on OpenStreetMap data using diffusion models.
- **Urban object counting (MMM 2025):** Developed methods to count unique objects across geo-tagged street images, with an application to homeless encampments in Los Angeles.

CV Researcher

Image Processing & Distributed Systems Lab, Shahid Beheshti University

Advisor: Prof. Mohsen Ebrahimi Moghadam

Sep 2022 – Feb 2023

Tehran, Iran

- Developed chaotic activation functions and feature extractors for a ConvLSTM model predicting respiratory motion in 4D CT scans for radiotherapy planning; undergraduate thesis published at IMPROVE 2024.

Machine Learning Intern

Institute for Research in Fundamental Sciences (IPM)

Jun 2021 – Sep 2021

Tehran, Iran

- Developed a hybrid CNN-LSTM model in Python and TensorFlow for stock price and trend prediction, combining multi-scale feature extraction with temporal modeling.

Technical Skills

Core: Python, PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, OpenCV, Git, AWS, Azure

Additional: C/C++, SQL, Java, MATLAB, C#, Go, JavaScript, HTML/CSS, Django, GeoPy, LaTeX

Selected Publications and Preprints

ORBIT: Training-Free Multi-Attribute Behavioral Steering via Orthogonal Subspace Rotation

Narges Ghasemi, Amir Ziashahabi, Salman Avestimehr, Jonathan May. *arXiv preprint*, 2026.

GeoToken: Hierarchical Geolocalization of Images via Next Token Prediction

Narges Ghasemi*, Amir Ziashahabi*, Salman Avestimehr, Cyrus Shahabi. *IEEE ICDM*, 2025.

OSMGen: Highly Controllable Satellite Image Synthesis using OpenStreetMap Data

Amir Ziashahabi*, Narges Ghasemi*, Sajjad Shahabi, John Krumm, Salman Avestimehr, Cyrus Shahabi. *NeurIPS UrbanAI Workshop*, 2025.

Assessment of Foot Clearance Impairments in Glaucoma Patients Using Smart Insoles

Cem Teker, Narges Ghasemi, et al. *International Symposium on Visual Computing (ISVC)*, 2025.

Counting Unique Objects in Geo-Tagged Street Images: A Case Study of Homeless Encampments in Los Angeles

Narges Ghasemi, Seon Ho Kim, Abdullah Alfarrarjeh, Cyrus Shahabi. *International Conference on Multimedia Modeling (MMM)*, 2025.

Chaotic Convolutional Long Short-Term Memory Network for Respiratory Motion Prediction

Narges Ghasemi, Shahabedin Nabavi, Mohsen Ebrahimi Moghadam, Yasser Shekofteh. *International Conference on Image Processing and Vision Engineering (IMPROVE)*, 2024.

* Equal contribution.

Additional Projects

GeoImage2Vec (*Python, PyTorch, contrastive learning*)

Combined visual and spatial features through contrastive learning for multimodal georeferenced image representations.

BysGNN for POI Forecasting (*Python, PyTorch, GNNs, LLMs*)

Combined graph neural networks with language-model semantic embeddings to predict point-of-interest visits.

Truck Detection and Counting (*Python, YOLO, DeepSORT, StrongSORT*)

Developed an object detection and tracking pipeline for urban traffic analysis.

Flash Reflection Removal (*Python, U-Net*)

Applied a U-Net model to remove flash reflections from images for an Image Processing course project.

Hamshahri News Classification (*Python, scikit-learn, SVM*)

Built a Persian news classifier using SVM and feature engineering; ranked 2nd in a Kaggle competition.

Teaching Experience

Teaching Assistant, University of Southern California

Machine Learning for Data Science; Introduction to Artificial Intelligence.

Teaching Assistant, Shahid Beheshti University

Machine Learning (graduate course); Software/Hardware Codesign; Computer Simulation; Artificial Intelligence; Operating Systems; Discrete Mathematics; Linear Algebra; Signal & Systems; Engineering Mathematics; Design and Analysis of Algorithms; Advanced Programming; Theory of Languages and Automata.

Academic Honors

- Ranked 1st among 131 undergraduate Computer Engineering students, 2018–2023.
- 18th place, Digikala Cup Data Science Competition (out of 1,380 competitors), 2021.
- 19th place, AUT ACM-ICPC Competition, 2019.
- Full-tuition scholarship; top 0.5% of 251K+ in the nationwide B.Sc. university entrance exam, 2018.
- Selected, First Round of Iranian Mathematics Olympiad, 2017.
- Semi-finalist, IMC (Selection Test of International Mathematics Olympiad), 2016.
- Bronze medal, AITMO (Selection Test of Asia Inter-Cities Teenagers Mathematics Olympiad), 2013.

Selected Courses

Advanced Natural Language Processing; Text as Data; Probabilistic and Generative Models; Machine Learning; Computational Intelligence; Image Processing.